IB MATHEMATICS SL • UNIT 7.01

Topic 7: AA SL Analytical Proofs, Exact Trig & Kinematics

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

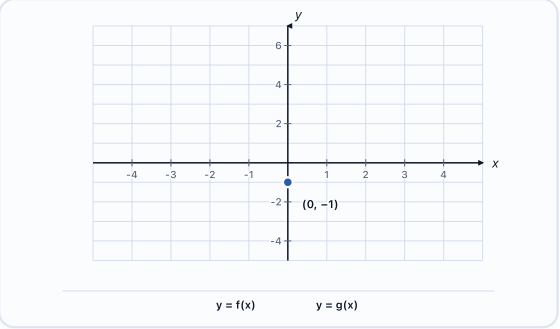
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 C	Q3 B	Q4 A	Q5 B
Q6 B	Q7 A	Q8 A	Q9 A	Q10 A
Q11 C	Q12 C	Q13 A	Q14 A	Q15 C
Q16 C	Q17 B	Q18 B	Q19 B	Q20 D
Q21 C	Q22 D	Q23 D	Q24 C	Q25 A

Topic 7: AA SL Analytical Proofs, Exact Trig & Kinematics Practice Exercises

1 MCQ

Consider the function $f(x) = x^2 - 5x + 6$. What is the value of $f(2)$?



A 0 ✓

B -2

C 4

D 8

WORKED SOLUTION & VERIFICATION

To find the value of $f(2)$, substitute $x = 2$ into the function $f(x) = x^2 - 5x + 6$.

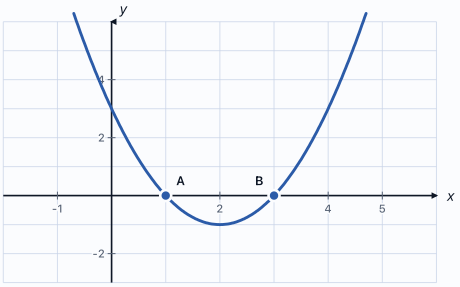
$f(2) = (2)^2 - 5(2) + 6$

$f(2) = 4 - 10 + 6$

$f(2) = 0$ **[M1] [A1]**

2 MCQ

Consider the quadratic function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. The points where the graph intersects the x-axis are denoted by A and B. What is the distance between points A and B?



A 1

B 3

C 2 ✓

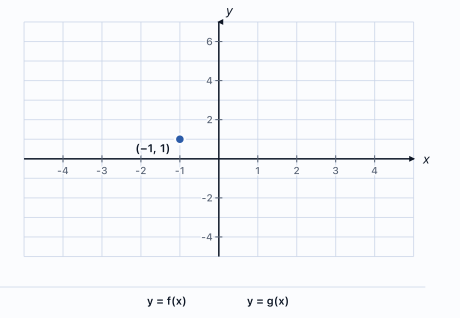
D 4

WORKED SOLUTION & VERIFICATION

The points where the graph intersects the x-axis are the roots of the quadratic equation $f(x) = 0$. We need to solve $x^2 - 4x + 3 = 0$. Factoring the quadratic, we get $(x - 1)(x - 3) = 0$. Thus, the roots are $x = 1$ and $x = 3$. Let point A have coordinates $(1, 0)$ and point B have coordinates $(3, 0)$. The distance between A and B is the absolute difference of their x-coordinates: $|3 - 1| = 2$. **[M1] [A1]**

3 MCQ

Consider the function $f(x) = x^2 - 2x + 1$. Determine the y-coordinate of the vertex of the graph of $y = f(x)$.



A 1

B 0 ✓

C -1

D 2

WORKED SOLUTION & VERIFICATION

The graph of a quadratic function $f(x) = ax^2 + bx + c$ is a parabola. The x-coordinate of the vertex is given by $x = -\frac{b}{2a}$. For the function $f(x) = x^2 - 2x + 1$, we have $a = 1$ and $b = -2$. Therefore, the x-coordinate of the vertex is $x = -\frac{-2}{2(1)} = \frac{2}{2} = 1$. To find the y-coordinate of the vertex, we substitute this value of x back into the function: $f(1) = (1)^2 - 2(1) + 1 = 1 - 2 + 1 = 0$. Thus, the y-coordinate of the vertex is 0. **[M1] [A1]**

4 MCQ

Consider the function $f(x) = 3x^2 - 2x + 5$. Find the value of $f'(1)$.

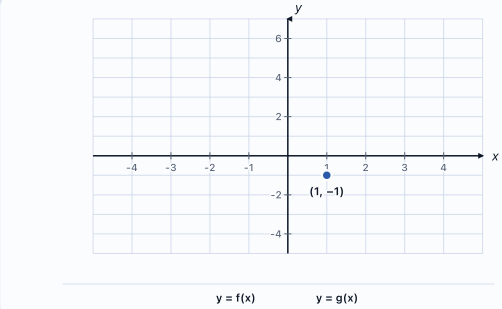
- **A** 4 ✓
- **B** 5
- **C** 6
- **D** 7

WORKED SOLUTION & VERIFICATION

The derivative of a function $f(x)$ gives the instantaneous rate of change of the function at any given point. We first find the derivative of $f(x) = 3x^2 - 2x + 5$ using the power rule: $f'(x) = \frac{d}{dx}(3x^2) - \frac{d}{dx}(2x) + \frac{d}{dx}(5)$. Applying the power rule, $\frac{d}{dx}(ax^n) = nax^{n-1}$ and the derivative of a constant is 0, we get $f'(x) = 2(3x^{2-1}) - 1(2x^{1-1}) + 0$, which simplifies to $f'(x) = 6x - 2$. To find the value of the derivative at $x = 1$, we substitute $x = 1$ into the expression for $f'(x)$: $f'(1) = 6(1) - 2 = 6 - 2 = 4$. **[M1] [A1]**

5 MCQ

The graph of the function $f(x) = 2x^2 - 3x + 5$ is shown below. Determine the value of $f'(2)$.



- **A** 7
- **B** 5 ✓
- **C** 9
- **D** 2

WORKED SOLUTION & VERIFICATION

The derivative of the function $f(x) = 2x^2 - 3x + 5$ is found using the power rule: $f'(x) = \frac{d}{dx}(2x^2) - \frac{d}{dx}(3x) + \frac{d}{dx}(5)$. This gives $f'(x) = 4x - 3$. To find $f'(2)$, substitute $x = 2$ into the derivative: $f'(2) = 4(2) - 3 = 8 - 3 = 5$. **[M1] [A1]**

6 MCQ

Consider the function $f(x) = x^2 - 6x + 5$. Find the x-coordinate of the vertex of the parabola represented by $y = f(x)$.

- **A** 1
- **B** 3 ✓
- **C** -1
- **D** -3

WORKED SOLUTION & VERIFICATION

The given function is a quadratic function in the form $f(x) = ax^2 + bx + c$, where $a = 1$, $b = -6$, and $c = 5$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values of a and b, we get $x = -\frac{-6}{2(1)} = \frac{6}{2} = 3$. Therefore, the x-coordinate of the vertex is 3. **[M1] [A1]**

7 MCQ

Find $\int x^2 + 4x^3\,dx$.

- **A** $\frac{1}{3}x^3 + x^4 + C$ ✓
- **B** $\frac{1}{4}x^4 + \frac{4}{5}x^5 + C$
- **C** $3x^3 + 16x^4 + C$
- **D** $x^2 + 4x^3 + C$

WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int x^2 + 4x^3\,dx = \frac{1}{3}x^3 + x^4 + C$. **[M1]**

[A1]

8 MCQ

Find $\int 5x^2 + 4x\,dx$.

- **A** $\frac{5}{3}x^3 + 2x^2 + C$ ✓
- **B** $\frac{5}{4}x^4 + \frac{4}{3}x^3 + C$
- **C** $15x^3 + 8x^2 + C$
- **D** $5x^2 + 4x + C$

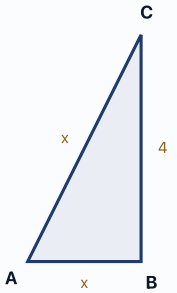
WORKED SOLUTION & VERIFICATION

Working through this calculus step: Using the power rule $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $\int 5x^2 + 4x\,dx = \frac{5}{3}x^3 + 2x^2 + C$. **[M1]**

[A1]

9 MCQ

In the diagram, triangle ABC is shown with coordinates A(1, 1), B(3, 1), and C(3, 5). What is the exact length of the side AC in simplest radical form?



- **A** $2\sqrt{5}$ ✓
- **B** 4
- **C** 5
- **D** $4\sqrt{2}$

WORKED SOLUTION & VERIFICATION

Using the distance formula between $A(1, 1)$ and $C(3, 5)$:

$$AC = \sqrt{(3-1)^2 + (5-1)^2} = \sqrt{2^2 + 4^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}.$$

[M1] [A1]

10 MCQ

Convert 210° to radians, giving your answer as an exact multiple of π.

- **A** $\frac{7\pi}{6}$ ✓
- **B** $\frac{5\pi}{3}$
- **C** $\frac{\pi}{2}$
- **D** $\frac{7\pi}{4}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $210^\circ \times \frac{\pi}{180} = \frac{7\pi}{6}$. **[M1] [A1]**

11 MCQ

Convert 270° to radians, giving your answer as an exact multiple of π.

- **A** $\frac{3\pi}{4}$
- **B** $\frac{\pi}{4}$
- **C** $\frac{3\pi}{2}$ ✓
- **D** $\frac{2\pi}{3}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $270^\circ \times \frac{\pi}{180} = \frac{3\pi}{2}$. **[M1] [A1]**

12 MCQ

Convert 240° to radians, giving your answer as an exact multiple of π.

- **A** $\frac{7\pi}{6}$
- **B** π
- **C** $\frac{4\pi}{3}$ ✓
- **D** $\frac{\pi}{3}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $240^\circ \times \frac{\pi}{180} = \frac{4\pi}{3}$. **[M1] [A1]**

13 MCQ

Convert 330° to radians, giving your answer as an exact multiple of π.

- **A** $\frac{11\pi}{6}$ ✓
- **B** $\frac{\pi}{2}$
- **C** $\frac{4\pi}{3}$
- **D** $\frac{\pi}{3}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $330^\circ \times \frac{\pi}{180} = \frac{11\pi}{6}$. **[M1] [A1]**

14 MCQ

Convert 300° to radians, giving your answer as an exact multiple of π.

- **A** $\frac{5\pi}{3}$ ✓
- **B** $\frac{\pi}{3}$
- **C** $\frac{2\pi}{3}$
- **D** $\frac{4\pi}{3}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $300^\circ \times \frac{\pi}{180} = \frac{5\pi}{3}$. **[M1] [A1]**

15 MCQ

Convert 120° to radians, giving your answer as an exact multiple of π.

- **A** $\frac{5\pi}{4}$
- **B** $\frac{7\pi}{6}$
- **C** $\frac{2\pi}{3}$ ✓
- **D** $\frac{3\pi}{2}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $120^\circ \times \frac{\pi}{180} = \frac{2\pi}{3}$. **[M1] [A1]**

16 MCQ

Convert 150° to radians, giving your answer as an exact multiple of π.

- **A** $\frac{3\pi}{4}$
- **B** $\frac{\pi}{2}$
- **C** $\frac{5\pi}{6}$ ✓
- **D** $\frac{7\pi}{4}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $150^\circ \times \frac{\pi}{180} = \frac{5\pi}{6}$. **[M1] [A1]**

17 MCQ

Convert 90° to radians, giving your answer as an exact multiple of π.

- **A** $\frac{5\pi}{6}$
- **B** $\frac{\pi}{2}$ ✓
- **C** $\frac{7\pi}{6}$
- **D** $\frac{2\pi}{3}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $90^\circ \times \frac{\pi}{180} = \frac{\pi}{2}$. **[M1] [A1]**

18 MCQ

Convert 45° to radians, giving your answer as an exact multiple of π.

A $\frac{11\pi}{6}$

B $\frac{\pi}{4}$ ✓

C π

D $\frac{5\pi}{3}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $45^\circ \times \frac{\pi}{180} = \frac{\pi}{4}$. **[M1] [A1]**

19 MCQ

Convert 225° to radians, giving your answer as an exact multiple of π.

A $\frac{11\pi}{6}$

B $\frac{5\pi}{4}$ ✓

C $\frac{\pi}{2}$

D $\frac{7\pi}{4}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $225^\circ \times \frac{\pi}{180} = \frac{5\pi}{4}$. **[M1] [A1]**

20 MCQ

Convert 315° to radians, giving your answer as an exact multiple of π.

A $\frac{\pi}{4}$

B $\frac{3\pi}{4}$

C $\frac{2\pi}{3}$

D $\frac{7\pi}{4}$ ✓

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $315^\circ \times \frac{\pi}{180} = \frac{7\pi}{4}$. **[M1] [A1]**

21 MCQ

Convert 60° to radians, giving your answer as an exact multiple of π.

A π

B $\frac{5\pi}{3}$

C $\frac{\pi}{3}$ ✓

D $\frac{5\pi}{6}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $60^\circ \times \frac{\pi}{180} = \frac{\pi}{3}$. **[M1] [A1]**

22 MCQ

Convert 135° to radians, giving your answer as an exact multiple of π.

A $\frac{\pi}{2}$

B $\frac{5\pi}{6}$

C $\frac{5\pi}{3}$

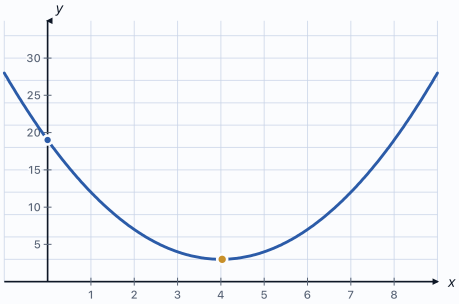
D $\frac{3\pi}{4}$ ✓

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $135^\circ \times \frac{\pi}{180} = \frac{3\pi}{4}$. **[M1] [A1]**

23 MCQ

In AA/AI SL — Functions, Statistics & Analytical Calculus (problem set #998), consider the function $f_{998}(x) = x^2 - 8x + 19$ illustrated on the coordinate axes. Express $f_{998}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{998}(6)$.



A 14

B 7

C 8

D 10 ✓

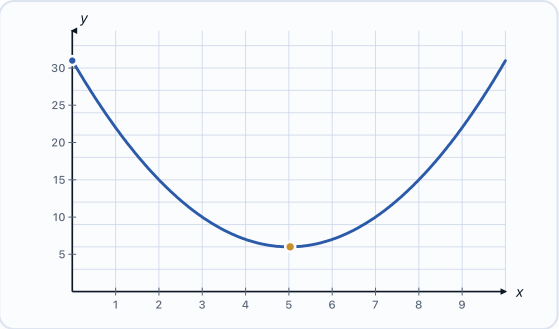
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (AA/AI SL — Functions, Statistics & Analytical Calculus):

1. Complete the square: $f_{998}(x) = (x-4)^2 + 3$, so $k = 3$. **[M1]**
2. Evaluate $f_{998}(6) = (6-4)^2 + 3 = 4 + 3 = 7$. **[M1]**
3. Sum: $k + f_{998}(6) = 3 + 7 = 10$. **[A1]**

24 MCQ

Considering AA/AI SL — Functions, Statistics & Analytical Calculus (problem set #1001), consider the function $f_{1001}(x) = x^2 - 10x + 31$ illustrated on the coordinate axes. Express $f_{1001}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{1001}(10)$.



A 31

B 32

C 37 ✓

D 42

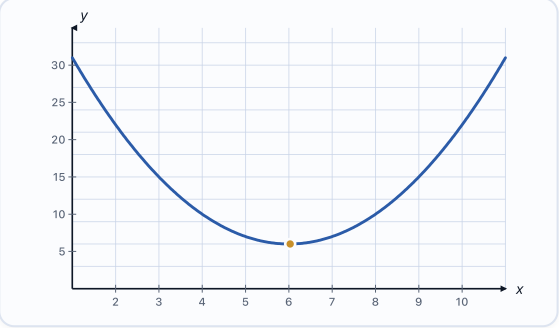
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (AA/AI SL — Functions, Statistics & Analytical Calculus):

1. Complete the square: $f_{1001}(x) = (x-5)^2 + 6$, so $k = 6$. **[M1]**
2. Evaluate $f_{1001}(10) = (10-5)^2 + 6 = 25 + 6 = 31$. **[M1]**
3. Sum: $k + f_{1001}(10) = 6 + 31 = 37$. **[A1]**

25 MCQ

Examining AA/AI SL — Functions, Statistics & Analytical Calculus (problem set #1004), consider the function $f_{1004}(x) = x^2 - 12x + 45$ illustrated on the coordinate axes. Express $f_{1004}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{1004}(14)$.



A 82 ✓

B 73

C 74

D 88

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (AA/AI SL — Functions, Statistics & Analytical Calculus):

1. Complete the square: $f_{1004}(x) = (x-6)^2 + 9$, so $k = 9$. **[M1]**
2. Evaluate $f_{1004}(14) = (14-6)^2 + 9 = 64 + 9 = 73$. **[M1]**
3. Sum: $k + f_{1004}(14) = 9 + 73 = 82$. **[A1]**